

COMPOSITE MATERIAL DATASHEETS GENERIC CFRP FOR TRADE-OFF STUDIES

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

MATERIAL IDENTIFICATION

Description	High performance IM carbon fibre reinforced UD-Tape material
Version	0.1 (26/11/2021)

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.184 mm
Density	1600 kg/m ³

Coefficients of Thermal Expansion CTE (1/°C)	α_1	α_2	α_3
	0.0	3E-5	3E-5

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12}	K_B
Values	155000	8500	4200	0.35	0.85

ALLOWABLE STRAINS

Material condition	Tension ϵ_{tall} (µε)	Compression ϵ_{call} (µε)
Pristine	15000	7000
Quality defects	12000	5500
Damage Tolerant	6000	3500

MATERIAL IDENTIFICATION

Description	HS carbon fibre reinforced UD-Tape material
Version	0.1 (26/11/2021)

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.184 mm
Density	1600 kg/m ³

Coefficients of Thermal Expansion CTE (1/°C)	α_1	α_2	α_3
	0.0	3E-5	3E-5

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12}	K_B
Values	135000	8500	4200	0.35	0.80

ALLOWABLE STRAINS

Material condition	Tension ϵ_{tall} (µε)	Compression ϵ_{call} (µε)
Pristine	10000	7000
Quality defects	8000	5500
Damage Tolerant	5000	3500